H0 — decision summary (1/2)

```
llama31-8b  (ctx 131,072,  1,024 heads)
    INPUT VALIDITY FAIL   haystack real
      ladder niah 4.044 b   cont 4.059 b   paired delta -0.010 b (need >= 0.1)
      heads with niah > cont  40.3%  (need >= 60%, n=1,024)
    HEADS IN BAND   69.1%  (gain over best corner >= 2.0x at 3 b/token)
    ROUTED GAIN     2.49x  (geometric mean of max(gain,1) over all heads)
    in-band heads  median   3.2x   p90   4.7x
    ladder width   median  4.06 b   IQR [3.39, 4.71]   100% of heads above 1.4
    tau            median  2.85   excl. sinks  2.85
    top-1 weight   median 0.0165   95% mass in 24906 tokens (20.7% of ctx)
    B=2b  vs uniform   16.0x   vs eviction    2.7x   vs BEST corner    2.6x   evicts 62%
    B=3b  vs uniform   15.8x   vs eviction    3.2x   vs BEST corner    2.8x   evicts 49%
    quantizer alpha @2b 0.941 (1.0 = no temperature distortion);  spearman@2b 0.606  [H2 answered]
    heads where eviction beats a 1-bit tier: 94%
    heads where eviction beats a 2-bit tier: 13%
    linearization check (predicted/measured gain @3b): 1.81  (1.0 = exact)
    VERDICT: UNKNOWN – input-validity gate failed (paired ladder delta -0.010 b < 0.1 b; only 40.3% of heads have niah
        > cont (need 60%)). The band fraction below is a measurement of this prompt, not of the model's long-context
        behaviour. See h0_measurement/bugs/1_from_synthetic_to_real_corpus/.

mistral-7b  (ctx 32,768,  1,024 heads)
    INPUT VALIDITY FAIL   haystack real
      ladder niah 2.675 b   cont 2.665 b   paired delta +0.002 b (need >= 0.1)
      heads with niah > cont  53.5%  (need >= 60%, n=1,024)
    HEADS IN BAND   84.5%  (gain over best corner >= 2.0x at 3 b/token)
    ROUTED GAIN     4.06x  (geometric mean of max(gain,1) over all heads)
    in-band heads  median   4.6x   p90  10.5x
    ladder width   median  2.68 b   IQR [2.32, 3.12]   99% of heads above 1.4
    tau            median  1.94   excl. sinks  1.94
    top-1 weight   median 0.0053   95% mass in 10422 tokens (34.5% of ctx)
    B=2b  vs uniform    6.2x   vs eviction    3.7x   vs BEST corner    3.1x   evicts 53%
    B=3b  vs uniform    6.9x   vs eviction    6.6x   vs BEST corner    4.2x   evicts 36%
    quantizer alpha @2b 0.940 (1.0 = no temperature distortion);  spearman@2b 0.824  [H2 answered]
    heads where eviction beats a 1-bit tier: 67%
    heads where eviction beats a 2-bit tier: 3%
    linearization check (predicted/measured gain @3b): 2.35  (1.0 = exact)
    VERDICT: UNKNOWN – input-validity gate failed (paired ladder delta +0.002 b < 0.1 b; only 53.5% of heads have niah
        > cont (need 60%)). The band fraction below is a measurement of this prompt, not of the model's long-context
        behaviour. See h0_measurement/bugs/1_from_synthetic_to_real_corpus/.

qwen15-moe-a2.7b  (ctx 32,768,  384 heads)
    INPUT VALIDITY FAIL   haystack real
      ladder niah 4.293 b   cont 4.321 b   paired delta +0.028 b (need >= 0.1)
      heads with niah > cont  71.1%  (need >= 60%, n=384)
    HEADS IN BAND   68.8%  (gain over best corner >= 2.0x at 3 b/token)
    ROUTED GAIN     2.58x  (geometric mean of max(gain,1) over all heads)
    in-band heads  median   3.4x   p90   5.3x
    ladder width   median  4.29 b   IQR [3.54, 5.33]   99% of heads above 1.4
    tau            median  3.04   excl. sinks  3.04
    top-1 weight   median 0.0172   95% mass in 7986 tokens (26.5% of ctx)
    B=2b  vs uniform    7.7x   vs eviction    3.0x   vs BEST corner    2.6x   evicts 61%
    B=3b  vs uniform    9.6x   vs eviction    3.9x   vs BEST corner    2.9x   evicts 47%
    quantizer alpha @2b 0.943 (1.0 = no temperature distortion);  spearman@2b 0.583  [H2 answered]
    heads where eviction beats a 1-bit tier: 84%
    heads where eviction beats a 2-bit tier: 7%
    linearization check (predicted/measured gain @3b): 2.42  (1.0 = exact)
    VERDICT: UNKNOWN – input-validity gate failed (paired ladder delta +0.028 b < 0.1 b). The band fraction below is a
        measurement of this prompt, not of the model's long-context behaviour. See
        h0_measurement/bugs/1_from_synthetic_to_real_corpus/.
```

H0 — decision summary (2/2)

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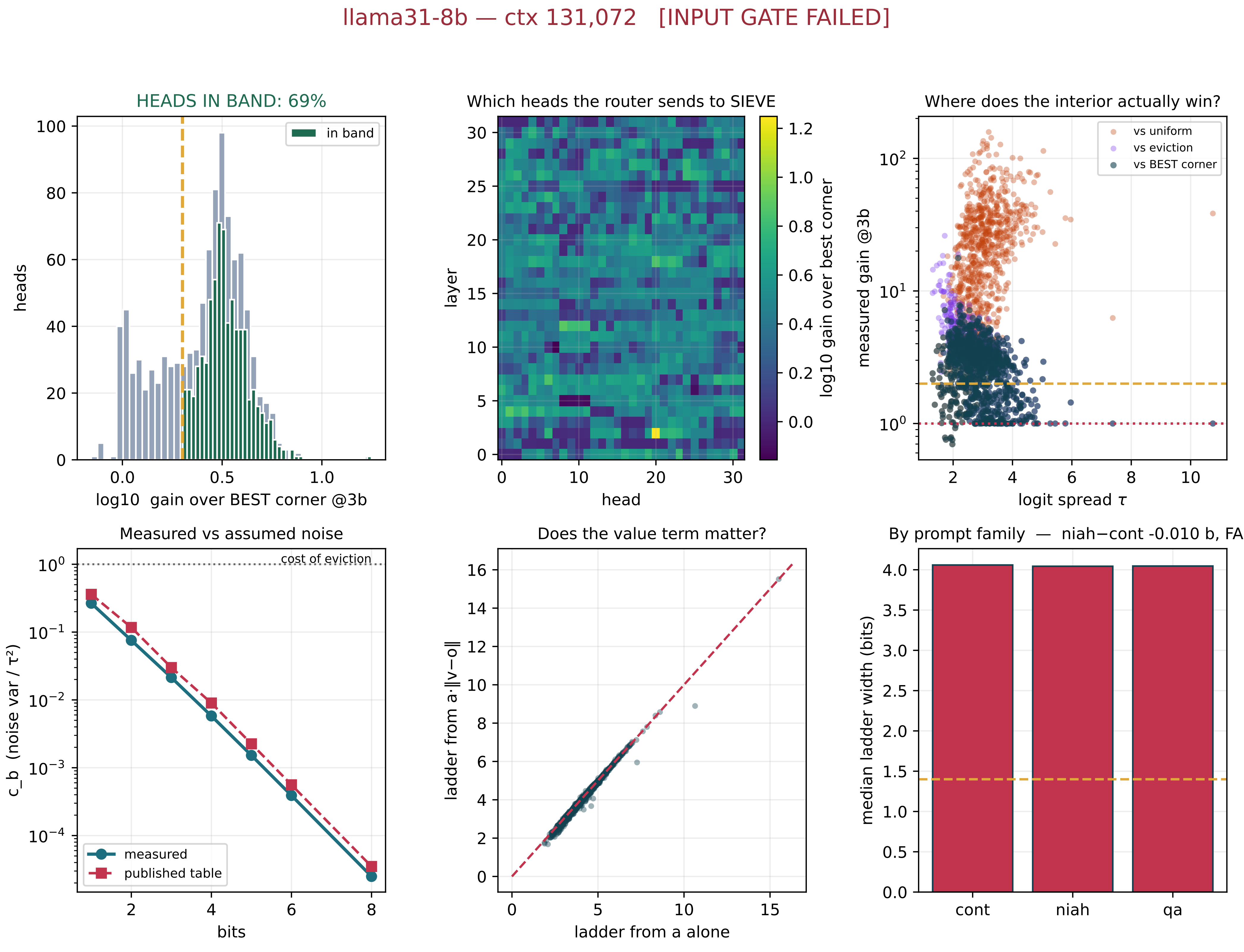
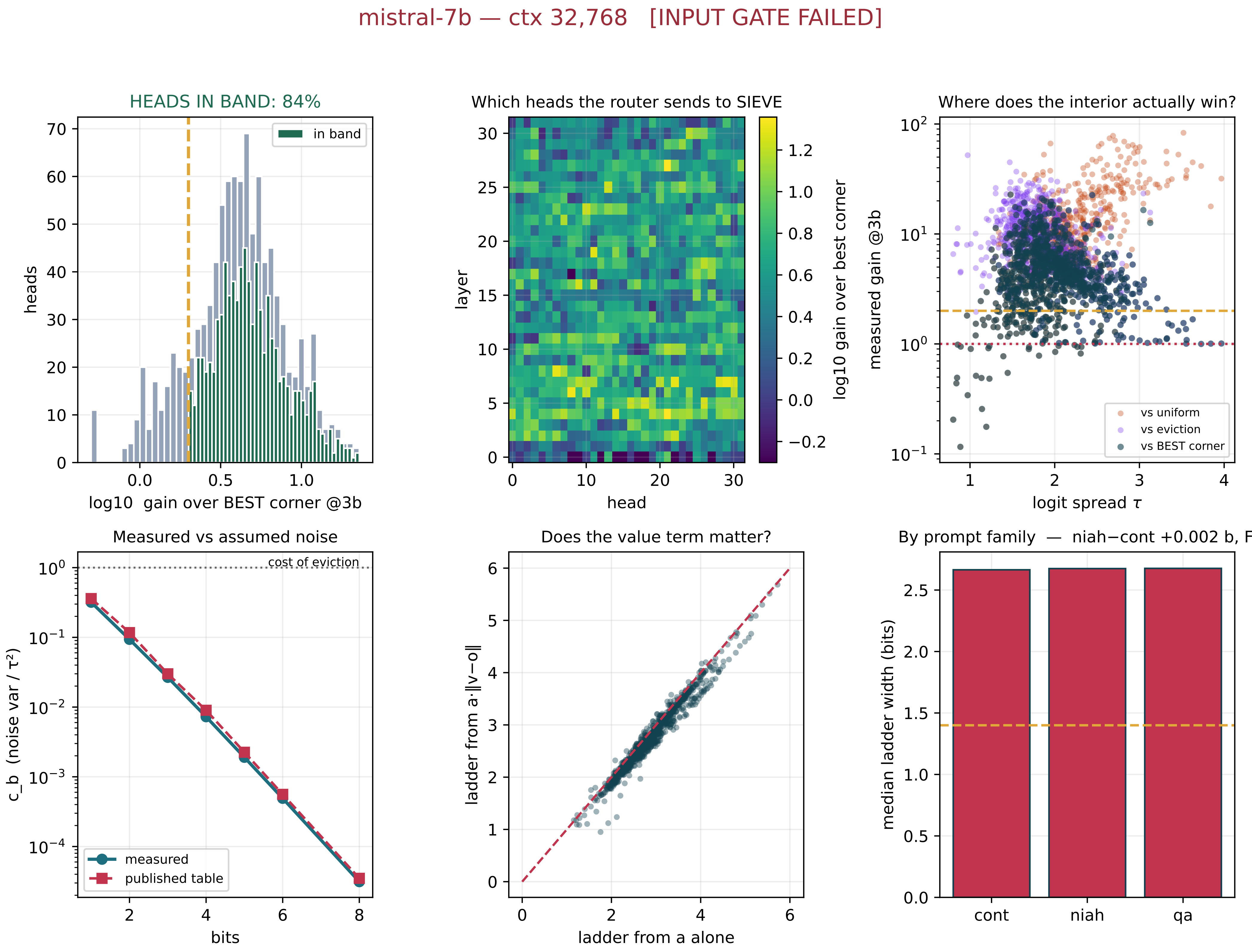
qwen3-30b-a3b-2507  (ctx 131,072,  1,536 heads)
    INPUT VALIDITY FAIL   haystack real
      ladder niah 4.603 b   cont 4.495 b   paired delta +0.061 b (need >= 0.1)
      heads with niah > cont  62.9%  (need >= 60%, n=1,536)
    HEADS IN BAND   14.1%  (gain over best corner >= 2.0x at 3 b/token)
    ROUTED GAIN     1.27x  (geometric mean of max(gain,1) over all heads)
    in-band heads  median   2.4x   p90   3.1x
    ladder width   median  4.55 b   IQR [3.93, 5.44]   100% of heads above 1.4
    tau            median  3.15   excl. sinks  3.15
    top-1 weight   median 0.1188   95% mass in 2424 tokens (2.0% of ctx)
    B=2b  vs uniform   59.7x   vs eviction    1.3x   vs BEST corner    1.3x   evicts 68%
    B=3b  vs uniform   33.7x   vs eviction    1.0x   vs BEST corner    1.0x   evicts 54%
    quantizer alpha @2b 0.941 (1.0 = no temperature distortion);  spearman@2b 0.493  [H2 answered]
    heads where eviction beats a 1-bit tier: 100%
    heads where eviction beats a 2-bit tier: 75%
    linearization check (predicted/measured gain @3b): 0.86  (1.0 = exact)
    VERDICT: UNKNOWN — input-validity gate failed (paired ladder delta +0.061 b < 0.1 b). The band fraction below is a
        measurement of this prompt, not of the model's long-context behaviour. See
        h0_measurement/bugs/1_from_synthetic_to_real_corpus/.

qwen3-8b  (ctx 40,960,  1,152 heads)
    INPUT VALIDITY FAIL   haystack real
      ladder niah 4.421 b   cont 4.441 b   paired delta -0.010 b (need >= 0.1)
      heads with niah > cont  43.1%  (need >= 60%, n=1,152)
    HEADS IN BAND   54.3%  (gain over best corner >= 2.0x at 3 b/token)
    ROUTED GAIN     1.94x  (geometric mean of max(gain,1) over all heads)
    in-band heads  median   2.9x   p90   3.9x
    ladder width   median  4.42 b   IQR [3.62, 5.92]   100% of heads above 1.4
    tau            median  3.15   excl. sinks  3.15
    top-1 weight   median 0.0150   95% mass in 6252 tokens (16.6% of ctx)
    B=2b  vs uniform   13.4x   vs eviction    2.1x   vs BEST corner    2.0x   evicts 64%
    B=3b  vs uniform   15.2x   vs eviction    2.5x   vs BEST corner    2.2x   evicts 50%
    quantizer alpha @2b 0.942 (1.0 = no temperature distortion);  spearman@2b 0.467  [H2 answered]
    heads where eviction beats a 1-bit tier: 99%
    heads where eviction beats a 2-bit tier: 39%
    linearization check (predicted/measured gain @3b): 1.75  (1.0 = exact)
    VERDICT: UNKNOWN — input-validity gate failed (paired ladder delta -0.010 b < 0.1 b; only 43.1% of heads have niah
        > cont (need 60%)). The band fraction below is a measurement of this prompt, not of the model's long-context
        behaviour. See h0_measurement/bugs/1_from_synthetic_to_real_corpus/.
```

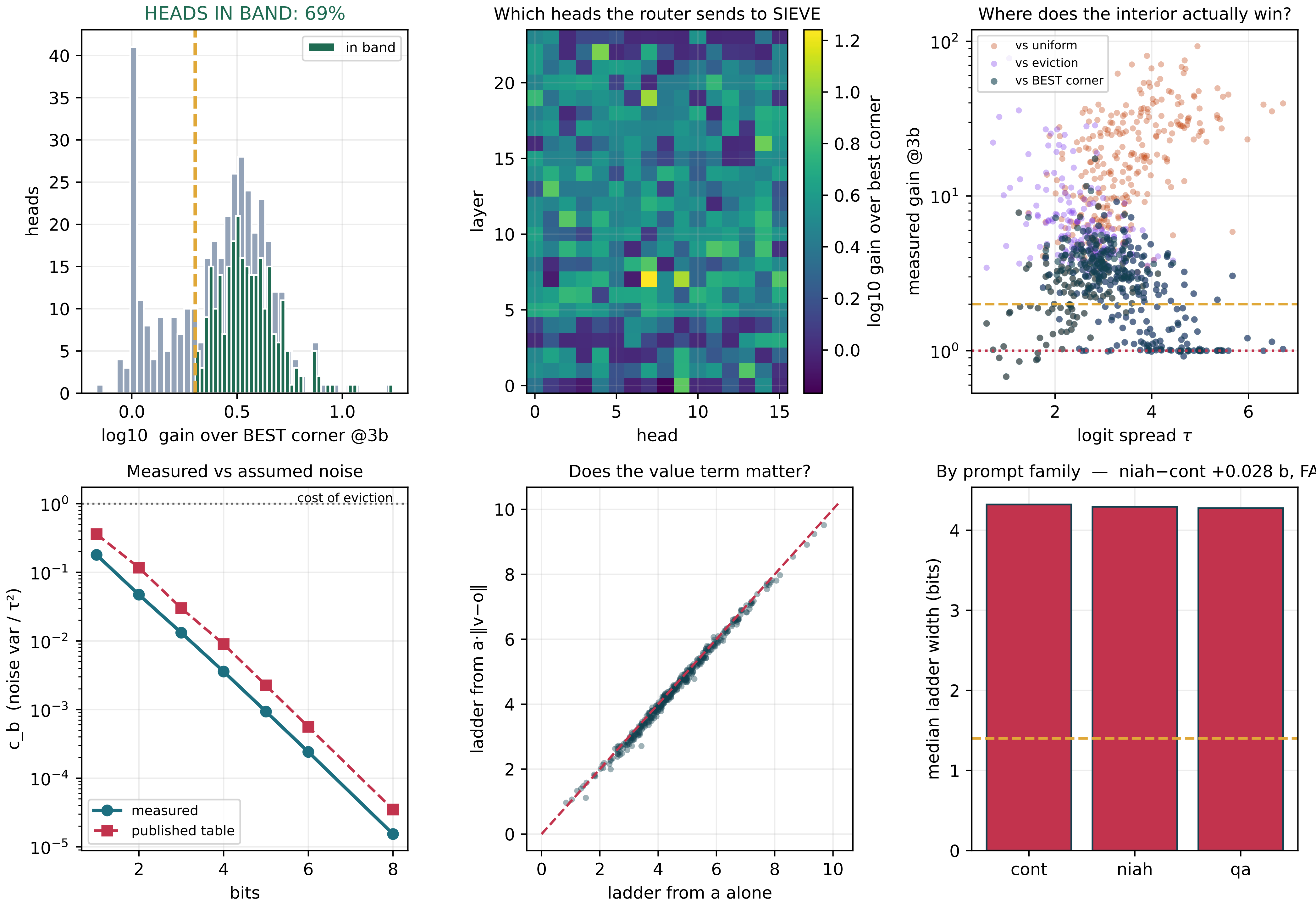


llama31-8b — ctx 131,072 [INPUT GATE FAILED]
HEADS IN BAND: 69%
in band
heads
log10 gain over BEST corner @3b
Which heads the router sends to SIEVE
layer
head
log10 gain over best corner
Where does the interior actually win?
vs uniform
vs eviction
vs BEST corner
measured gain @3b
logit spread τ
Measured vs assumed noise
cost of eviction
c_b (noise var / τ²)
measured
published table
bits
Does the value term matter?
ladder from a·‖v−o‖
ladder from a alone
By prompt family — niah−cont -0.010 b, FA
median ladder width (bits)
cont
niah
qa

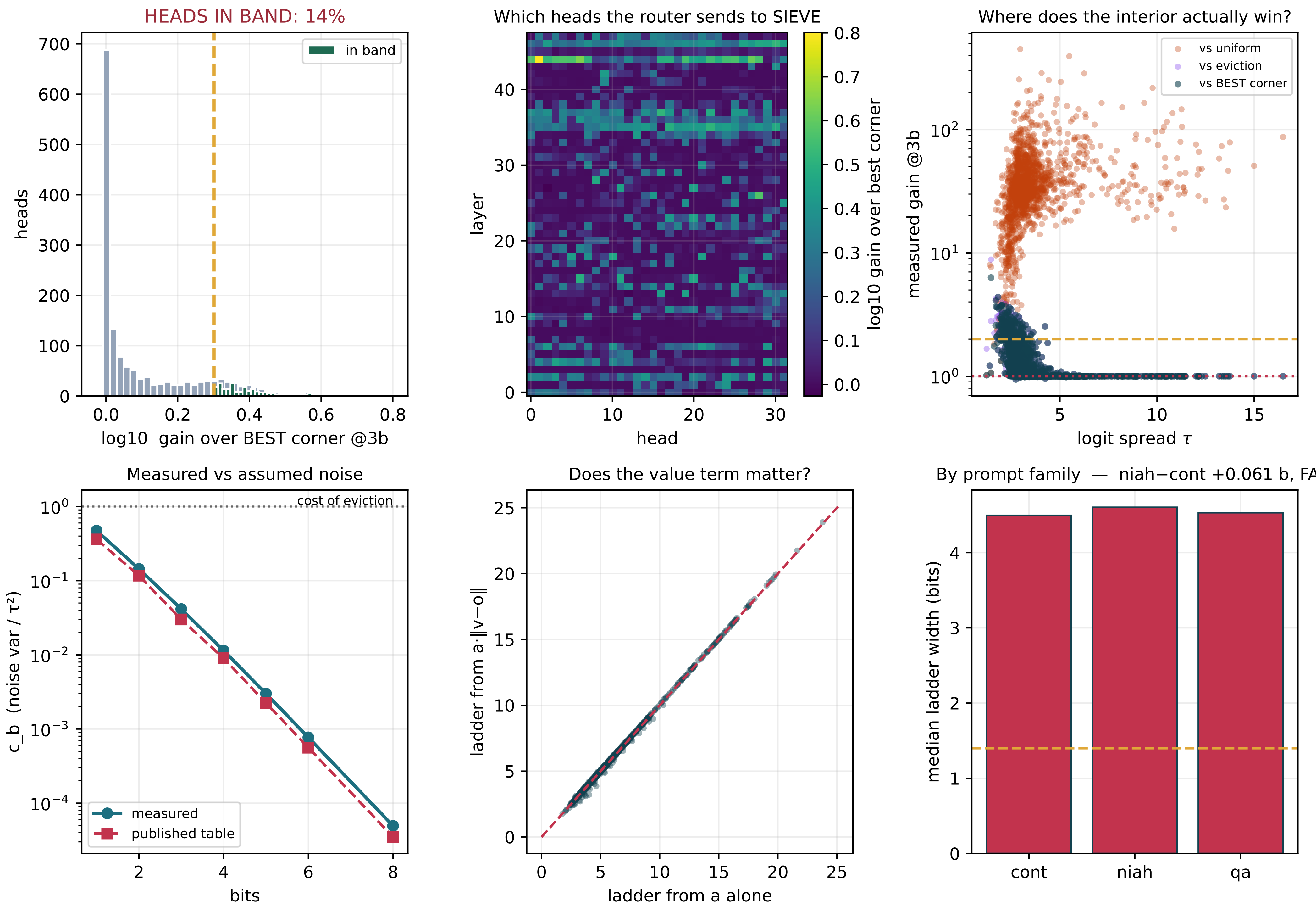


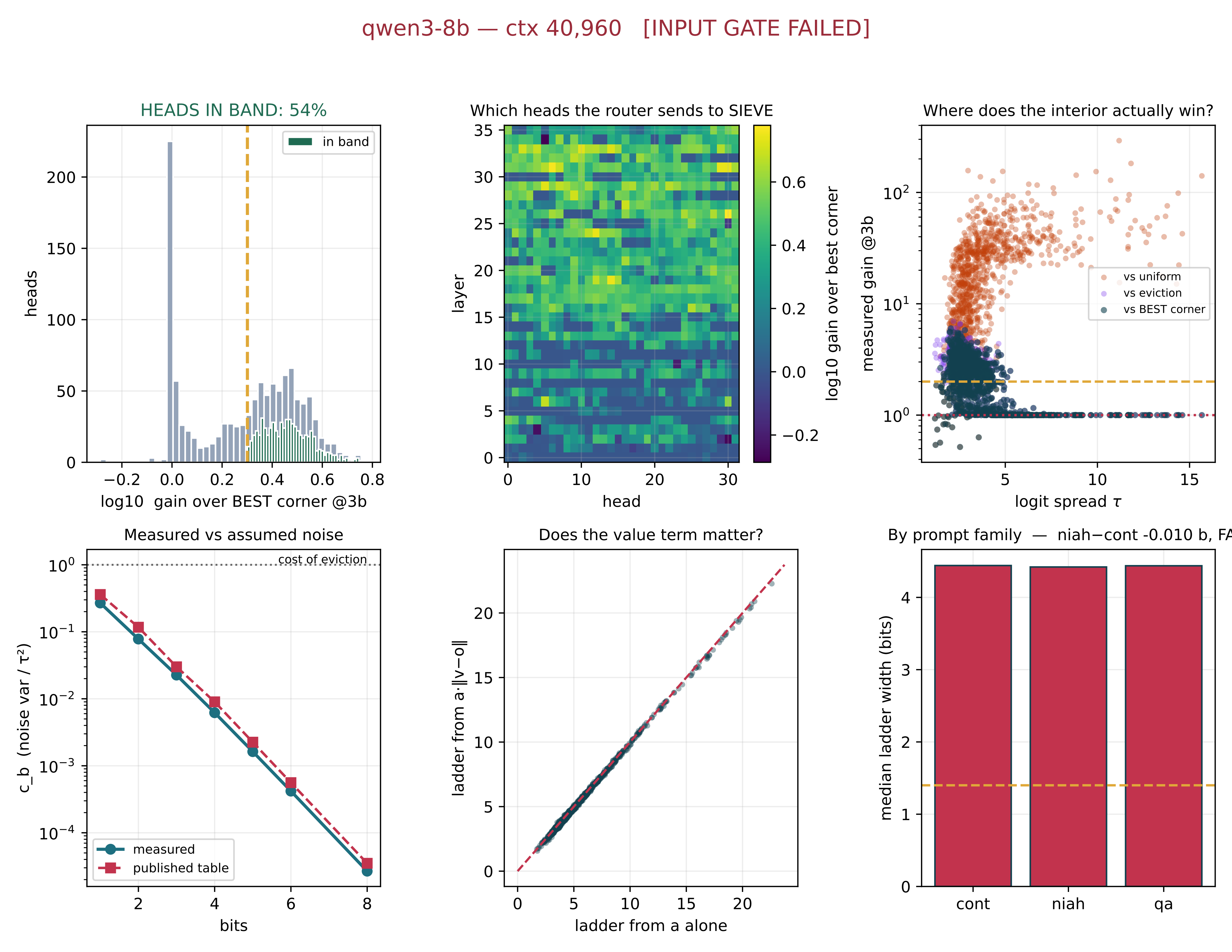
mistral-7b — ctx 32,768 [INPUT GATE FAILED]
HEADS IN BAND: 84%
in band
heads
log10 gain over BEST corner @3b
Which heads the router sends to SIEVE
layer
head
log10 gain over best corner
Where does the interior actually win?
measured gain @3b
logit spread τ
vs uniform
vs eviction
vs BEST corner
Measured vs assumed noise
cost of eviction
c_b (noise var / τ²)
bits
measured
published table
Does the value term matter?
ladder from a·‖v−o‖
ladder from a alone
By prompt family — niah−cont +0.002 b, F
median ladder width (bits)
cont
niah
qa

qwen15-moe-a2.7b — ctx 32,768   [INPUT GATE FAILED]



qwen3-30b-a3b-2507 — ctx 131,072 [INPUT GATE FAILED]





qwen3-8b — ctx 40,960 [INPUT GATE FAILED]
HEADS IN BAND: 54%
in band
heads
log10 gain over BEST corner @3b
Which heads the router sends to SIEVE
layer
head
log10 gain over best corner
Where does the interior actually win?
vs uniform
vs eviction
vs BEST corner
measured gain @3b
logit spread τ
Measured vs assumed noise
cost of eviction
measured
published table
c_b (noise var / τ²)
bits
Does the value term matter?
ladder from a·‖v−o‖
ladder from a alone
By prompt family — niah−cont -0.010 b, FA
median ladder width (bits)
cont
niah
qa

Cross-model comparison